An algorithm for SOI encounter computation on Keplerian orbits

Artur Sinila

Astro Graphic Rust Research Foundation

March 2026

Contents

1. Problem Statement

Given two Keplerian orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI}. It is possible that such E_1, E_2 don't exist, the algorithm needs to handle this case as well. The spacecraft orbit may be elliptic ($e_1 < 1$) or hyperbolic ($e_1 > 1$); the planet orbit is always elliptic.

$$\|\boldsymbol{A}\boldsymbol{r}_1(E_1) - \boldsymbol{B}\boldsymbol{r}_2(E_2)\| = r_{\text{SOI}} \tag{1}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

We divide this hard problem into two easier ones:
1. Finding tight **ranges** of E_1, E_2 **where** encounters are **geometrically** possible, ignoring real positions of the bodies
2. Finding precise **values** of E_1 and time t **when** the spacecraft **actually** enters sphere-of-influence of the planet

2. Position in Orbital Plane

For an elliptic orbit, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\boldsymbol{r}_i(E_i) = \begin{bmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{bmatrix} \tag{2}$$

For a hyperbolic spacecraft orbit:

$$\boldsymbol{r}_1(H_1) = \begin{bmatrix} a_1(\cosh H_1 - e_1) \\ b_1 \sinh H_1 \end{bmatrix} \tag{3}$$

3. Distance Constraint

Square both sides of the constraint:

$$(\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2)^\top(\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2) = r_{\text{SOI}}^2 \tag{4}$$

Expand the left-hand side:

$$\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{r}_1 - 2\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{B}\boldsymbol{r}_2 + \boldsymbol{r}_2^\top \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{r}_2 = r_{\text{SOI}}^2 \tag{5}$$

3.1. Orthonormality of $\boldsymbol{A}$ and $\boldsymbol{B}$

Since $\boldsymbol{A}$ and $\boldsymbol{B}$ are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}_{2\times 2}, \quad \boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_{2\times 2} \tag{6}$$

Therefore the self-terms simplify:

$$\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{r}_1 = \boldsymbol{r}_1^\top \boldsymbol{r}_1 = |\boldsymbol{r}_1|^2 =: r_1^2 \tag{7}$$

$$\boldsymbol{r}_2^\top \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{r}_2 = \boldsymbol{r}_2^\top \boldsymbol{r}_2 = |\boldsymbol{r}_2|^2 =: r_2^2 \tag{8}$$

3.2. Coupling matrix $\boldsymbol{C}$

Define the 2×2 matrix:

$$\boldsymbol{C} := 2\boldsymbol{A}^\top \boldsymbol{B} \tag{9}$$

This encodes the entire mutual orientation of the two orbital planes. Equation 5 becomes:

$$r_1^2 + r_2^2 - \boldsymbol{r}_1^\top \boldsymbol{C}\boldsymbol{r}_2 = r_{\text{SOI}}^2 \tag{10}$$

Note: $\boldsymbol{C}$ is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \le 2$.

3.3. Scaled coupling matrix $\boldsymbol{M}$

The position vector Equation 2 can be factored as $\boldsymbol{r}_i = \operatorname{diag}(a_i, b_i)\boldsymbol{p}_i$, where $\operatorname{diag}(a_i, b_i) := \begin{pmatrix} a_i & 0 \\ 0 & b_i \end{pmatrix}$ is a diagonal matrix and $\boldsymbol{p}_i$ is a dimensionless position-direction vector:

$$\boldsymbol{p}_i = \begin{bmatrix} \cos E_i - e_i \\ \sin E_i \end{bmatrix} \tag{11}$$

For a hyperbolic spacecraft orbit:

$$\boldsymbol{p}_1 = \begin{bmatrix} \cosh H_1 - e_1 \\ \sinh H_1 \end{bmatrix} \tag{12}$$

Substituting $\boldsymbol{r}_i = \operatorname{diag}(a_i, b_i)\boldsymbol{p}_i$ into the cross-term and using $(\boldsymbol{X}\boldsymbol{p})^\top = \boldsymbol{p}^\top \boldsymbol{X}^\top$:

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = (\operatorname{diag}(a_1, b_1)\boldsymbol{p}_1)^\top \boldsymbol{C} \operatorname{diag}(a_2, b_2)\boldsymbol{p}_2 = \boldsymbol{p}_1^\top \operatorname{diag}(a_1, b_1)^\top \boldsymbol{C} \operatorname{diag}(a_2, b_2)\boldsymbol{p}_2 \tag{13}$$

Since diagonal matrices are symmetric $(\operatorname{diag}(x, y)^\top = \operatorname{diag}(x, y))$:

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \boldsymbol{p}_1^\top \operatorname{diag}(a_1, b_1)\boldsymbol{C} \operatorname{diag}(a_2, b_2)\boldsymbol{p}_2 \tag{14}$$

We have a chain of five matrices. Applying associativity ($\boldsymbol{A}(\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{A}\boldsymbol{B})\boldsymbol{C}$) twice – first to $(\boldsymbol{p}_1^\top \cdot \operatorname{diag}(a_1, b_1)) \cdot \boldsymbol{C} = \boldsymbol{p}_1^\top \cdot (\operatorname{diag}(a_1, b_1) \cdot \boldsymbol{C})$, then to $(\boldsymbol{p}_1^\top \cdot (\operatorname{diag}(a_1, b_1) \cdot \boldsymbol{C})) \cdot \operatorname{diag}(a_2, b_2) = \boldsymbol{p}_1^\top \cdot ((\operatorname{diag}(a_1, b_1) \cdot \boldsymbol{C}) \cdot \operatorname{diag}(a_2, b_2))$ – lets us group the three constant middle factors into a single **scaled coupling matrix $\boldsymbol{M}$**.

$$\boldsymbol{M} = \operatorname{diag}(a_1, b_1)\boldsymbol{C} \operatorname{diag}(a_2, b_2) \tag{15}$$

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{16}$$

Let's introduce matrix $\boldsymbol{D}$ and express $\boldsymbol{M}$ in terms of it ($\circ$ denotes element-wise product):

$$\boldsymbol{D} := \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}^\top = \begin{pmatrix} a_1 a_2 & a_1 b_2 \\ b_1 a_2 & b_1 b_2 \end{pmatrix}, \quad \boldsymbol{M} := \boldsymbol{D} \circ \boldsymbol{C} \tag{17}$$

4. Coarse encounter intervals

We restrict the search to eccentric anomaly values where the spacecraft's heliocentric distance lies between planetary perihelion and aphelion $\pm r_{\mathrm{SOI}}$.

4.1. Radial overlap condition

An encounter requires the spacecraft distance to lie within:

$$a_2(1 - e_2) - r_{\mathrm{SOI}} \le r_1 \le a_2(1 + e_2) + r_{\mathrm{SOI}} \tag{18}$$

Elliptic spacecraft. Using $r_1 = a_1(1 - e_1 \cos E_1)$ and solving for $\cos E_1$:

$$\frac{a_1 - a_2(1 + e_2) - r_{\mathrm{SOI}}}{a_1 e_1} \le \cos E_1 \le \frac{a_1 - a_2(1 - e_2) + r_{\mathrm{SOI}}}{a_1 e_1} \tag{19}$$

Clamping both sides to $[-1, 1]$ and applying arccos (which reverses the inequality) gives two symmetric intervals $[E_{\mathrm{lo}}, E_{\mathrm{hi}}]$ and $[-E_{\mathrm{lo}}, -E_{\mathrm{hi}}]$ in E_1. If no valid interval exists (i.e. the clamped range is empty), no encounter is possible at any E_1.

Hyperbolic spacecraft. Using $r_1 = a_1(1 - e_1 \cosh H_1)$ and solving for $\cosh H_1$ (noting $a_1 e_1 < 0$, so dividing reverses the inequality):

$$\frac{a_1 - a_2(1 - e_2) + r_{\mathrm{SOI}}}{a_1 e_1} \le \cosh H_1 \le \frac{a_1 - a_2(1 + e_2) - r_{\mathrm{SOI}}}{a_1 e_1} \tag{20}$$

Clamping the lower bound to $\max(..., 1)$ (since $\cosh H_1 \ge 1$) and applying arcosh gives a single symmetric interval $[-H_{\mathrm{hi}}, -H_{\mathrm{lo}}] \cup [H_{\mathrm{lo}}, H_{\mathrm{hi}}]$. If the clamped range is empty, no encounter is possible.

4.2. Initial guess via midpoint and atan2 projection

Newton's method (Section 5) converges rapidly once seeded near a local minimum of f. We produce the seed in two cheap steps: take the midpoint of the radial-overlap interval as E_1^*, then estimate the matching E_2^* via a closed-form atan2 projection.

4.2.1. Selecting E_1^* as the interval midpoint

The radial-overlap condition Equation 19 produces the symmetric intervals $[E_{\text{lo}}, E_{\text{hi}}]$ and $[-E_{\text{hi}}, -E_{\text{lo}}]$. For each branch we seed Newton's method at the midpoint:

$$E_1^* = (E_{\text{lo}} + E_{\text{hi}})/2 \tag{21}$$

No f evaluations are spent on the seed itself — the interval from the radial-overlap condition already brackets the region where encounters are geometrically possible, and its midpoint is within half the interval width of the true minimum. Newton's quadratic convergence absorbs the residual in a handful of iterations.

4.2.2. Estimating E_2 from E_1 via atan2 projection

Given E_1^*, the spacecraft's position in the ecliptic frame is $\boldsymbol{A}\boldsymbol{r}_1(E_1^*)$. Project it onto the planet's orbital plane and express the result in the planet's 2D coordinates:

$$\boldsymbol{q} := \boldsymbol{B}^\top \boldsymbol{A}\boldsymbol{r}_1(E_1^*) \tag{22}$$

We now want an E_2 such that $\boldsymbol{r}_2(E_2)$ is as close as possible to $\boldsymbol{q}$. Finding the **exact** closest-point-on-ellipse has no closed form below degree 4 (Section 4.3 notes the same quartic structure), so we look for a cheap approximation derived from the ellipse's natural parametrisation.

4.2.2.1. Deriving atan2 projection from an affine transformation

The ellipse has Sun (focus) at the origin and **centre** at $(-a_2 e_2, 0)$, with equation

$$(x + a_2 e_2)^2 / a_2^2 + y^2 / b_2^2 = 1. \tag{23}$$

Define the affine map

$$T : (x, y) \mapsto (x/a_2 + e_2, \quad y/b_2). \tag{24}$$

T shifts the ellipse centre onto the new origin (via $+a_2 e_2$ in x after the $1/a_2$ scaling) and rescales both axes so the ellipse equation becomes

$$x'^2 + y'^2 = 1, \tag{25}$$

a **unit circle centred at the new origin**. The Sun, which was at the original origin, is carried by T to $(e_2, 0)$ in the new frame — it is no longer at the origin.

The parametrisation maps particularly cleanly. Applying T to $\boldsymbol{r}_2(E_2) = (a_2(\cos E_2 - e_2), b_2 \sin E_2)$:

$$T(\boldsymbol{r}_2(E_2)) = (\cos E_2, \quad \sin E_2). \tag{26}$$

So in the T-frame, E_2 is exactly the angular coordinate on the unit circle measured from its centre (which is also the new origin). This is the geometric meaning of eccentric anomaly, hiding in plain sight in the parametrisation.

Closest-point-on-unit-circle is now a genuinely solvable problem. Let $(p_x, p_y) := T(\boldsymbol{q}) = (q_x/a_2 + e_2, q_y/b_2)$. A point on the unit circle is parametrised by E_2 as $(\cos E_2, \sin E_2)$. We seek the E_2 minimising

$$D^2(E_2) := (\cos E_2 - p_x)^2 + (\sin E_2 - p_y)^2 \tag{27}$$

Differentiating:

$$\frac{dD^2}{dE_2} = 2(p_x \sin E_2 - p_y \cos E_2) \tag{28}$$

Setting $dD^2/dE_2 = 0$:

$$p_x \sin E_2 = p_y \cos E_2 \quad \Longrightarrow \quad E_2 = \operatorname{atan2}(p_y, p_x) \tag{29}$$

The second derivative $d^2 D^2 / dE_2^2 = 2(p_x \cos E_2 + p_y \sin E_2)$ evaluated at this root is $2\sqrt{p_x^2 + p_y^2} > 0$, confirming it is a minimum (the other stationary point, at $E_2 + \pi$, is the maximum). Substituting the definitions of p_x, p_y:

$$E_2 = \operatorname{atan2}(q_y/b_2, \quad q_x/a_2 + e_2) \tag{30}$$

The derivation implicitly assumes $(p_x, p_y) \neq (0, 0)$, or equivalently $\boldsymbol{q} \neq (-a_2 e_2, 0)$. This degenerate case occurs only when $\boldsymbol{q}$ sits exactly at the ellipse centre, far inside the orbit — not geometrically relevant for encounter problems, where $\boldsymbol{q}$ is on the order of the planet's heliocentric distance.

4.2.2.2. Why this works (and where it approximates)

When the two orbital planes coincide ($i_{\text{mut}} = 0$), $\boldsymbol{q}$ lies on the planet's ellipse, $T(\boldsymbol{q})$ lies on the unit circle, and the closest-point-on-circle is $T(\boldsymbol{q})$ itself: Equation 30 returns the exact E_2.

For inclined orbits, the orthogonal projection $\boldsymbol{B}^\top \boldsymbol{A} \boldsymbol{r}_1$ shortens $\boldsymbol{q}$ relative to its true heliocentric extent, so $T(\boldsymbol{q})$ moves off the unit circle and Equation 30 returns only the **angle** of the closest circle point, not the true minimiser of the Euclidean distance on the original ellipse (the affine T stretches the metric non-uniformly). This is the structural source of the approximation error.

Figure 1 shows both frames. A naïve alternative — "pick the ellipse point along the Sun-ray through $\boldsymbol{q}$" — is strictly worse: the focal ray is not perpendicular to the ellipse at the intersection because the focus is off-centre, so that intersection is not a closest point. Only the affine transformation T, which places the **ellipse centre** (not the Sun) at the new origin, makes ray-from-origin geometrically meaningful.

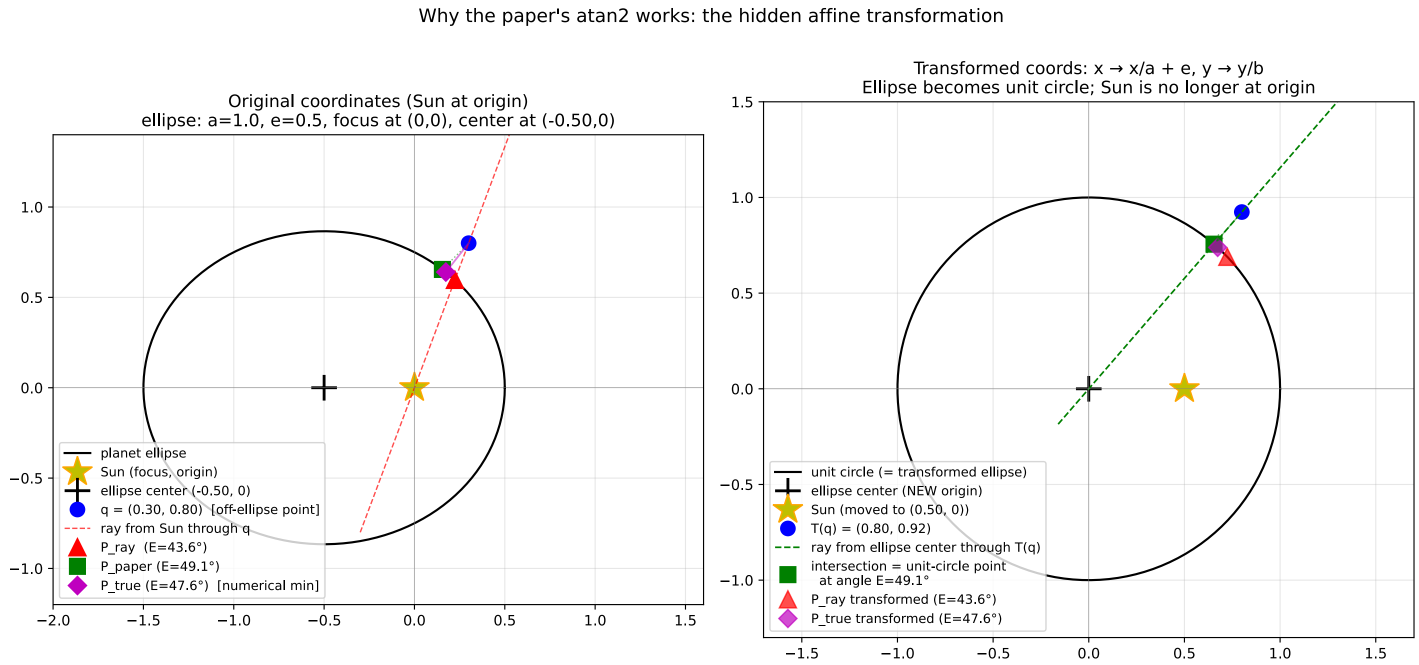

Figure 1: Geometric derivation of the atan2 projection, for an ellipse with $a = 1$, $e = 0.5$. **Left:** original coordinates; Sun (yellow star) at the focus = origin, ellipse centre at $(-ae, 0)$ is a distinct point. An off-ellipse probe point $\boldsymbol{q}$ (blue dot) has three candidate eccentric anomalies: the Sun-ray intersection (red triangle), the atan2 projection Equation 30 (green square), and the true minimum-distance point (magenta diamond, numerical). **Right:** the affine transformation T maps the ellipse to a unit circle. The new origin coincides with the **ellipse centre**; the Sun has moved to $(e, 0)$. The atan2 projection is now simply the ray from the new origin through $T(\boldsymbol{q})$, hitting the unit circle at the closest circle point. Back-transforming recovers the green square.

4.2.2.3. Accuracy

When the orbital planes coincide, the projection is exact and Equation 30 returns the correct E_2. For inclined orbits, the error grows with the mutual inclination i_{mut} and with the spacecraft eccentricity e_1 (which spreads the spacecraft trajectory further from the planet's plane). Numerical evaluation for a Pluto-like target ($i_{\text{mut}} = 17°$, $e_2 = 0.25$) shows the maximum E_2 error is under $2°$ for moderate $e_1 < 0.5$, growing to $\sim 9°$ at favourable Hohmann-like geometries ($e_1 \approx 0.95$) and up to $\sim 63°$ at unfavourable ascending-node orientations. Newton's method converges reliably from any of these seeds, so the formula is adequate despite its worst-case degradation.

4.3. Initial guess via mutual line of nodes

An alternative initial guess exploits the geometry of the two orbital planes directly. The **mutual line of nodes** is the intersection of the two planes — a line through the common focus (the Sun). Its direction is:

$$\ell = \hat{\boldsymbol{n}}_1 \times \hat{\boldsymbol{n}}_2, \quad \hat{\boldsymbol{n}}_1 = \boldsymbol{A}_{:,1} \times \boldsymbol{A}_{:,2}, \quad \hat{\boldsymbol{n}}_2 = \boldsymbol{B}_{:,1} \times \boldsymbol{B}_{:,2} \tag{31}$$

where $\hat{\boldsymbol{n}}_1$ is the unit normal to the spacecraft's orbital plane and $\hat{\boldsymbol{n}}_2$ is the unit normal to the planet's orbital plane (each obtained as the cross product of the two orthonormal columns of $\boldsymbol{A}$ and $\boldsymbol{B}$ respectively). Hence $|\ell| = \sin(i_{\text{mut}})$ with i_{mut} the mutual inclination between the two planes. When both bodies lie on the same ray from the Sun along ℓ, the out-of-plane component of their separation vanishes — this is a natural 3D minimum-distance configuration.

Project ℓ into each orbital plane: $\boldsymbol{\lambda}_1 = \boldsymbol{A}^\top \ell$, $\boldsymbol{\lambda}_2 = \boldsymbol{B}^\top \ell$ (both 2D). The eccentric anomaly at which $\boldsymbol{r}_i(E_i)$ is parallel to $\boldsymbol{\lambda}_i$ satisfies the cross-product equation $(\boldsymbol{r}_i \times \boldsymbol{\lambda}_i)_z = 0$:

$$a_i \lambda_y \cos E_i - b_i \lambda_x \sin E_i = a_i e_i \lambda_y \tag{32}$$

which using $A \cos E - B \sin E = R \cos(E - \phi)$ with $R = \sqrt{(a_i \lambda_y)^2 + (b_i \lambda_x)^2}$, $\phi = \operatorname{atan2}(-b_i \lambda_x, a_i \lambda_y)$ reduces to:

$$E_i = \phi \pm \arccos\left(\frac{a_i e_i \lambda_y}{R}\right) \tag{33}$$

The two roots correspond to $\boldsymbol{r}_i \parallel +\ell$ and $\boldsymbol{r}_i \parallel -\ell$; classify by the sign of $\boldsymbol{r}_i(E_i) \cdot \boldsymbol{\lambda}_i$. The **same-side** pairings (E_1^+, E_2^+) and (E_1^-, E_2^-) are the two candidate initial guesses.

Hyperbolic spacecraft. Equation 32 becomes $A \cosh H - B \sinh H = a_1 e_1 \lambda_y$. When $A^2 > B^2$, set $R = \operatorname{sgn}(A)\sqrt{A^2 - B^2}$, $\phi = \operatorname{atanh}(B/A)$, giving $H_1 = \phi \pm \operatorname{arcosh}(a_1 e_1 \lambda_y / R)$ whenever $a_1 e_1 \lambda_y / R \geq 1$. If the discriminant or arcosh argument fails, the line of nodes does not intersect the reachable branch of the hyperbola.

Experimental comparison. We evaluated both initial-guess strategies against four Voyager 2 encounters — three elliptic (Earth, Mars, Jupiter) and one hyperbolic (Saturn, using the Jupiter-to-Saturn leg) — and report the 3D separation d at the guess alongside the separation d^* at the converged true minimum.

Target	i_{mut}	d^* **(km)**	d_{atan2} **(km)**	d_{nod} **(km)**	$d_{\text{atan2}}/d_{\text{nod}}$
Earth (ell.)	5.02°	2.57×10^6	3.17×10^6	4.21×10^6	0.75
Mars (ell.)	5.11°	1.38×10^7	1.56×10^7	5.24×10^7	0.30
Jupiter (ell.)	5.99°	2.59×10^5	1.01×10^7	2.00×10^6	**5.05**
Saturn (hyp.)	0.26°	6.45×10^5	1.24×10^7	1.29×10^8	0.097

Table 1: Initial-guess 3D separation for the midpoint-plus-atan2 (d_{atan2}) and mutual-nodal (d_{nod}) strategies, versus the converged true minimum d^*. Earth/Mars are non-encounters (Voyager 2 does not enter their SOIs) and serve only as geometric probes. Jupiter is the intended close encounter, where the nodal guess wins by $5.05\times$. Saturn at $i_{\text{mut}} = 0.26°$ is a worst case for the nodal method: the line of nodes is poorly defined and the nodal root lands on the wrong asymptotic branch of the hyperbola.

Table 1 supports a simple dispatch rule: compute both guesses (each is $\mathcal{O}(1)$), evaluate f at each, and seed Newton with whichever yields smaller d.

5. Finding minimum distance using Newton's Method

We seek to minimise $f(E_1, E_2) \coloneqq r_1^2 + r_2^2 - \boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 - r_{\text{SOI}}^2$ using Newton's method. Each iteration solves the 2×2 linear system $\boldsymbol{H}\boldsymbol{\delta} = -\nabla f$ for the step $\boldsymbol{\delta} = (\delta E_1, \delta E_2)^\top$, then updates $E_i \leftarrow E_i + \delta E_i$.

5.1. Deriving the gradient and Hessian

We differentiate each term with respect to E_i.

5.1.1. Self-terms r_i^2

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$\frac{\partial r_i^2}{\partial E_i} = 2a_i^2 e_i \sin E_i (1 - e_i \cos E_i) \tag{34}$$

$$\frac{\partial^2 r_i^2}{\partial E_i^2} = 2a_i^2 e_i (\cos E_i - e_i + 2e_i \sin^2 E_i) \tag{35}$$

and $\partial r_i^2/\partial E_j = 0$ for $i \neq j$.

For a hyperbolic spacecraft orbit, $r_1 = a_1(1 - e_1 \cosh H_1)$ (positive since $a_1 < 0$ and $1 - e_1 \cosh H_1 < 0$):

$$\frac{\partial r_1^2}{\partial H_1} = -2a_1^2 e_1 \sinh H_1 (1 - e_1 \cosh H_1) \tag{36}$$

$$\frac{\partial^2 r_1^2}{\partial H_1^2} = 2a_1^2 e_1 (e_1 - \cosh H_1 + 2e_1 \sinh^2 H_1) \tag{37}$$

5.1.2. Cross-term $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2$

Since $\boldsymbol{M}$ is constant, differentiation acts only on $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$. Define the derivative vectors:

$$\hat{\boldsymbol{w}}_i := \mathrm{d}\frac{\boldsymbol{p}_i}{\mathrm{d}E_i} = \begin{bmatrix} -\sin E_i \\ \cos E_i \end{bmatrix}, \quad \hat{\boldsymbol{u}}_i := \begin{bmatrix} \cos E_i \\ \sin E_i \end{bmatrix} \tag{38}$$

For a hyperbolic spacecraft orbit:

$$\hat{\boldsymbol{w}}_1 := \mathrm{d}\frac{\boldsymbol{p}_1}{\mathrm{d}H_1} = \begin{bmatrix} \sinh H_1 \\ \cosh H_1 \end{bmatrix}, \quad \hat{\boldsymbol{u}}_1 := \begin{bmatrix} \cosh H_1 \\ \sinh H_1 \end{bmatrix} \tag{39}$$

To differentiate $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2$ with respect to E_1, note that only $\boldsymbol{p}_1$ depends on E_1. By associativity, $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 = \boldsymbol{p}_1^\top (\boldsymbol{M} \boldsymbol{p}_2)$, which is a dot product of $\boldsymbol{p}_1$ with the constant vector $\boldsymbol{M}\boldsymbol{p}_2$. The derivative of a dot product with one constant factor is:

$$\frac{\partial(\boldsymbol{p}_1^\top (\boldsymbol{M}\boldsymbol{p}_2))}{\partial E_1} = \hat{\boldsymbol{w}}_1^\top (\boldsymbol{M}\boldsymbol{p}_2) = \hat{\boldsymbol{w}}_1^\top \boldsymbol{M}\boldsymbol{p}_2 \tag{40}$$

Similarly, for E_2 only $\boldsymbol{p}_2$ depends on E_2. By associativity, $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 = (\boldsymbol{p}_1^\top \boldsymbol{M}) \boldsymbol{p}_2$, a dot product of the constant row vector $\boldsymbol{p}_1^\top \boldsymbol{M}$ with $\boldsymbol{p}_2$:

$$\frac{\partial((\boldsymbol{p}_1^\top \boldsymbol{M})\boldsymbol{p}_2)}{\partial E_2} = (\boldsymbol{p}_1^\top \boldsymbol{M})\hat{\boldsymbol{w}}_2 = \boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2 \tag{41}$$

The same reasoning applies to higher derivatives, replacing $\boldsymbol{p}_i$ by successive derivatives $\hat{\boldsymbol{w}}_i$, $-\hat{\boldsymbol{u}}_i$ (elliptic) or $+\hat{\boldsymbol{u}}_1$ (hyperbolic, since $\mathrm{d}^2\boldsymbol{p}_1/\,\mathrm{d}H_1^2 = +\hat{\boldsymbol{u}}_1$):

$$\frac{\partial^2(\boldsymbol{p}_1^\top \boldsymbol{M}\boldsymbol{p}_2)}{\partial E_1^2} = -\hat{\boldsymbol{u}}_1^\top \boldsymbol{M}\boldsymbol{p}_2, \quad \frac{\partial^2(\boldsymbol{p}_1^\top \boldsymbol{M}\boldsymbol{p}_2)}{\partial E_2^2} = -\boldsymbol{p}_1^\top \boldsymbol{M}\hat{\boldsymbol{u}}_2 \tag{42}$$

For a hyperbolic spacecraft orbit, the sign of the E_1 term flips:

$$\frac{\partial^2(\boldsymbol{p}_1^\top \boldsymbol{M}\boldsymbol{p}_2)}{\partial H_1^2} = +\hat{\boldsymbol{u}}_1^\top \boldsymbol{M}\boldsymbol{p}_2 \tag{43}$$

$$\frac{\partial^2(\boldsymbol{p}_1^\top \boldsymbol{M}\boldsymbol{p}_2)}{\partial E_1 \partial E_2} = \hat{\boldsymbol{w}}_1^\top \boldsymbol{M}\hat{\boldsymbol{w}}_2 \tag{44}$$

5.1.3. Gradient

$$\frac{\partial f}{\partial E_1} = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\boldsymbol{w}}_1^\top \boldsymbol{M}\boldsymbol{p}_2 \tag{45}$$

$$\frac{\partial f}{\partial E_2} = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \boldsymbol{p}_1^\top \boldsymbol{M}\hat{\boldsymbol{w}}_2 \tag{46}$$

For a hyperbolic spacecraft orbit, Equation 45 becomes (using Equation 36 and the hyperbolic $\hat{\boldsymbol{w}}_1$, $\boldsymbol{p}_1$ from Equation 39):

$$\frac{\partial f}{\partial H_1} = -2a_1^2 e_1 \sinh H_1 (1 - e_1 \cosh H_1) - \hat{\boldsymbol{w}}_1^\top \boldsymbol{M}\boldsymbol{p}_2 \tag{47}$$

Equation 46 is unchanged (the planet is always elliptic), but $\boldsymbol{p}_1$ and $\boldsymbol{M}$ use the hyperbolic definitions.

5.1.4. Hessian

The Hessian $\boldsymbol{H}$ is symmetric ($H_{12} = H_{21}$):

$$H_{11} = 2a_1^2 e_1 (\cos E_1 - e_1 + 2e_1 \sin^2 E_1) + \hat{\boldsymbol{u}}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{48}$$

$$H_{22} = 2a_2^2 e_2 (\cos E_2 - e_2 + 2e_2 \sin^2 E_2) + \boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{u}}_2 \tag{49}$$

$$H_{12} = -\hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2 \tag{50}$$

For a hyperbolic spacecraft orbit, H_{11} changes (using Equation 37 and Equation 43):

$$H_{11} = 2a_1^2 e_1 (e_1 - \cosh H_1 + 2e_1 \sinh^2 H_1) - \hat{\boldsymbol{u}}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{51}$$

H_{22} and H_{12} retain the same form, with the hyperbolic $\hat{\boldsymbol{w}}_1, \hat{\boldsymbol{u}}_1, \boldsymbol{p}_1$.

5.2. Solving the Newton step

The 2×2 system is solved directly via Cramer's rule:

$$\Delta = H_{11} H_{22} - H_{12}^2 \tag{52}$$

$$\delta E_1 = \frac{\frac{\partial f}{\partial E_1} H_{22} - \frac{\partial f}{\partial E_2} H_{12}}{-\Delta}, \quad \delta E_2 = \frac{\frac{\partial f}{\partial E_2} H_{11} - \frac{\partial f}{\partial E_1} H_{12}}{-\Delta} \tag{53}$$

All coupling terms are bilinear forms $\boldsymbol{x}^\top \boldsymbol{M} \boldsymbol{y}$ with 2D vectors, each requiring 4 multiplies and 3 adds. The five needed dot products ($\hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \boldsymbol{p}_2$, $\boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2$, $\hat{\boldsymbol{u}}_1^\top \boldsymbol{M} \boldsymbol{p}_2$, $\boldsymbol{p}_1^\top \boldsymbol{M} \hat{\boldsymbol{u}}_2$, $\hat{\boldsymbol{w}}_1^\top \boldsymbol{M} \hat{\boldsymbol{w}}_2$) can share the intermediate products $\boldsymbol{M}\boldsymbol{p}_2$, $\boldsymbol{M}\hat{\boldsymbol{w}}_2$, $\boldsymbol{M}\hat{\boldsymbol{u}}_2$ (each a 2D matrix-vector multiply).

6. Time Constraint via Kepler's Equation

The preceding sections treated E_1 and E_2 as independent variables. In reality, both bodies obey Kepler's equation, which couples each eccentric anomaly to a common time t.

6.1. Kepler's equation

The orbital period is $T_i = 2\pi \sqrt{|a_i|^3 / \mu}$, which applies to both elliptic and hyperbolic orbits since the derivation depends only on $|a_i|$.

For an elliptic orbit i with mean anomaly at epoch $M_{i,0}$:

$$M_i(t) = M_{i,0} + \frac{2\pi}{T_i} t = E_i - e_i \sin E_i \tag{54}$$

For a hyperbolic spacecraft orbit:

$$M_1(t) = M_{1,0} + \frac{2\pi}{T_1} t = e_1 \sinh H_1 - H_1 \tag{55}$$

At a given time t, both anomalies are determined:

$$E_1 - e_1 \sin E_1 = M_{1,0} + \frac{2\pi}{T_1} t, \quad E_2 - e_2 \sin E_2 = M_{2,0} + \frac{2\pi}{T_2} t \tag{56}$$

(replacing the first equation with Equation 55 for a hyperbolic spacecraft).

6.2. Eliminating time

The position on an ellipse is 2π-periodic in E_i, so each body returns to the same point after each full orbit. The set of times at which elliptic orbit i passes through eccentric anomaly $E_i \in [-\pi, \pi]$ is:

$$t = \frac{T_i}{2\pi} (E_i - e_i \sin E_i - M_{i,0} + 2\pi k_i) = \frac{T_i}{2\pi} (M_i - M_{i,0} + 2\pi k_i), \quad k_i \in \mathbb{Z} \tag{57}$$

where $M_i = E_i - e_i \sin E_i$ is the mean anomaly. A hyperbolic orbit is not periodic — the spacecraft passes through each H_1 exactly once:

$$t = \frac{T_1}{2\pi}\left(e_1 \sinh H_1 - H_1 - M_{1,0}\right) \tag{58}$$

For an encounter, both bodies must be at their respective positions **simultaneously**. For two elliptic orbits, equating the time expressions:

$$\frac{T_1}{2\pi}\left(M_1 - M_{1,0} + 2\pi k_1\right) = \frac{T_2}{2\pi}\left(M_2 - M_{2,0} + 2\pi k_2\right) \tag{59}$$

Simplifying and rearranging gives a family of **time-coupling constraints**, one for each integer pair (k_1, k_2):

$$h(E_1, E_2; \alpha) \coloneqq T_1\left(M_1 - M_{1,0}\right) - T_2\left(M_2 - M_{2,0}\right) + 2\pi\alpha = 0 \tag{60}$$

where:

$$\alpha = T_1 k_1 - T_2 k_2 \tag{61}$$

The parameter α has units of time: $T_1 k_1$ is the time for body 1 to complete k_1 orbits, $T_2 k_2$ is the time for body 2 to complete k_2 orbits, and α is the mismatch. Each choice of (k_1, k_2) corresponds to a different encounter opportunity (i.e. orbit 1 on its k_1-th revolution meeting orbit 2 on its k_2-th revolution).

In the (M_1, M_2) plane, Equation 60 defines a straight line with slope T_1 / T_2:

$$M_2 = \frac{T_1}{T_2}\left(M_1 - M_{1,0}\right) + M_{2,0} + \frac{2\pi\alpha}{T_2} \tag{62}$$

Different values of α shift the line parallel to itself. The full encounter problem is: find (E_1, E_2) satisfying both the distance constraint $f = 0$ (Equation 10) and $h = 0$ (Equation 60) for some admissible α.

Hyperbolic spacecraft. Since $k_1 = 0$ (Equation 58), the time constraint simplifies to a family parametrised by k_2 alone, with $\alpha = -T_2 k_2$:

$$h(H_1, E_2; k_2) \coloneqq T_1\left(e_1 \sinh H_1 - H_1 - M_{1,0}\right) - T_2\left(E_2 - e_2 \sin E_2 - M_{2,0}\right) - 2\pi T_2 k_2 = 0 \tag{63}$$

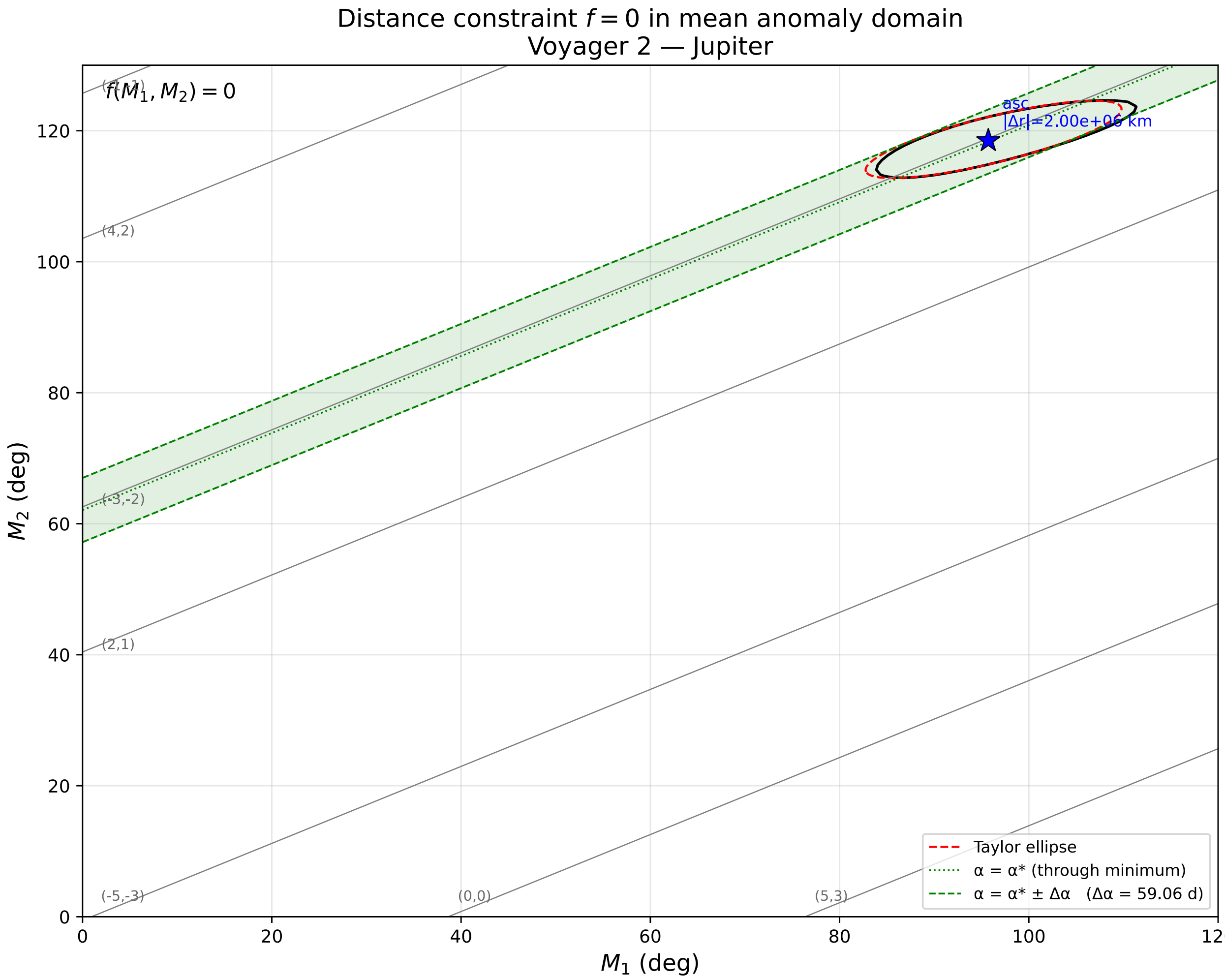

Figure 2: Distance constraint $f = 0$ and time-coupling lines for the Voyager 2 — Jupiter encounter, plotted in the (M_1, M_2) mean-anomaly domain. The black closed curve is the locus where the spacecraft–planet separation equals r_{SOI}; the red dashed ellipse is the second-order Taylor expansion of f about its minimum. Gray lines are the family of time constraints Equation 60, labelled by (k_1, k_2). An encounter exists wherever a gray line intersects the black curve. Blue stars mark the mutual-node same-side pairings (Section 4.3): the **ascending** pair lands at $(M_1, M_2) \approx (96°, 118°)$ with spacecraft–planet separation 2.0×10^6 km $\approx 0.04 r_{\text{SOI}}$ (inside the SOI, as expected for this encounter), while the **descending** pair at $(-2°, -52°)$ has separation 6.1×10^8 km $\approx 12.6 r_{\text{SOI}}$ (orbits pass far apart on the opposite node).

7. Finding the actual encounter

The geometric minimisation of Section 5 yields (E_1^*, E_2^*) where f attains its minimum $f^* \leq 0$ — a **potential** encounter ignoring when the bodies are actually there. The time constraint Equation 60 selects a discrete family of straight lines in (M_1, M_2), one per integer pair (k_1, k_2). The actual SOI entry is the earliest point (in t) lying on both the $f = 0$ contour and one of these lines.

7.1. Initial guess via line–ellipse intersection

Near (E_1^*, E_2^*), the second-order expansion of f defines an ellipse in E-space, $(\boldsymbol{E} - \boldsymbol{E}^*)^\top \boldsymbol{H}(\boldsymbol{E} - \boldsymbol{E}^*) \leq -2f^*$, where $\boldsymbol{H}$ is the Hessian from Section 5.1. Since the time constraint lives in (M_1, M_2), transform the Hessian into the M-domain using the diagonal Jacobian $\boldsymbol{J} = \operatorname{diag}(1 - e_1 \cos E_1^*, 1 - e_2 \cos E_2^*)$:

$$\boldsymbol{H}_M = \boldsymbol{J}^{-1} \boldsymbol{H} \boldsymbol{J}^{-1}, \quad (\boldsymbol{H}_M)_{ij} = \frac{H_{ij}}{(1 - e_i \cos E_i^*)\left(1 - e_j \cos E_j^*\right)} \tag{64}$$

The Taylor ellipse in (M_1, M_2) is then $(\boldsymbol{M} - \boldsymbol{M}^*)^\top \boldsymbol{H}_M (\boldsymbol{M} - \boldsymbol{M}^*) \leq -2f^*$.

Intersect the time line with this ellipse — a closed-form 2×2 quadratic in a single parameter. Take the intersection point with the smaller M_1 (earliest in time in this encounter zone). Convert back to eccentric anomalies by inverting Kepler's equation $M_i = E_i - e_i \sin E_i$ (one Newton iteration from $E_i \approx M_i$ is sufficient near the minimum).

7.2. Candidate selection on the integer lattice

Define α^* as the value of α that places the time line through the geometric minimum. Evaluating Equation 60 at (E_1^*, E_2^*):

$$\alpha^* := \frac{T_2\left(M_2^* - M_{2,0}\right) - T_1\left(M_1^* - M_{1,0}\right)}{2\pi}, \quad M_i^* := E_i^* - e_i \sin E_i^* \tag{65}$$

Admissible α lie on the lattice $\mathcal{L} := \{T_1 k_1 - T_2 k_2 : (k_1, k_2) \in \mathbb{Z}^2\}$. For each integer k_1, the k_2 that minimises $|\alpha - \alpha^*|$ is:

$$k_2(k_1) = \mathrm{round}\left(\frac{T_1 k_1 - \alpha^*}{T_2}\right). \tag{66}$$

With k_2 determined by k_1 this way, the search reduces to finding the smallest k_1 — subject to a mission-start floor — that puts the time line inside the Taylor ellipse around (M_1^*, M_2^*). The mission-start floor is

$$k_1^{\min} = \left\lceil t_{\mathrm{start}}/T_1 - \left(M_1^* - M_{1,0}\right)/(2\pi)\right\rceil, \tag{67}$$

ensuring $t_{\mathrm{enc}} \geq t_{\mathrm{start}}$ for the returned encounter. The remaining task is to locate the smallest $k_1 \geq k_1^{\min}$ whose $(k_1, k_2(k_1))$ satisfies $|\alpha - \alpha^*| \leq \Delta\alpha$, where $\Delta\alpha$ is the admissibility half-width derived next.

7.2.1. Admissibility window $\Delta\alpha$

As α varies, the time line slides without tilting — the slope $s := T_1/T_2$ is fixed, only the intercept moves. We want the largest drift of α from α^* before the line loses contact with the Taylor ellipse. At that drift, the line is **tangent** to the ellipse; call it $\Delta\alpha$.

Work in local coordinates $\Delta M_i := M_i - M_i^*$ with Hessian entries abbreviated $H_{ij}^M := (\boldsymbol{H}_M)_{ij}$. The ellipse is

$$H_{11}^M \Delta M_1^2 + 2H_{12}^M \Delta M_1 \Delta M_2 + H_{22}^M \Delta M_2^2 = -2f^*, \tag{68}$$

and the time line is $\Delta M_2 = s\,\Delta M_1 + d_0$ with $d_0 := (2\pi/T_2)(\alpha - \alpha^*)$.

Substitute the line into the ellipse. Plug $\Delta M_2 = s\Delta M_1 + d_0$ into Equation 68 and expand, collecting powers of ΔM_1:

$$A\Delta M_1^2 + B\Delta M_1 + C = 0, \tag{69}$$

where

$$\begin{aligned} A &:= H_{11}^M + 2s\,H_{12}^M + s^2 H_{22}^M, \\ B &:= 2d_0\left(H_{12}^M + s\,H_{22}^M\right), \\ C &:= d_0^2 H_{22}^M + 2f^*. \end{aligned} \tag{70}$$

Equation 69 has two real roots (line crosses ellipse at two points), one real double root (tangent, single intersection), or no real roots (line misses ellipse), depending on the sign of its discriminant.

Tangency = zero discriminant. The largest admissible $|d_0|$ is the value at which the line is tangent, i.e. $B^2 - 4AC = 0$:

$$4d_0^2\left(H_{12}^M + s\,H_{22}^M\right)^2 - 4A\left(d_0^2 H_{22}^M + 2f^*\right) = 0. \tag{71}$$

Dividing by 4 and isolating the d_0^2 coefficient:

$$d_0^2\left[\left(H_{12}^M + s\,H_{22}^M\right)^2 - A\,H_{22}^M\right] = 2Af^*. \tag{72}$$

Simplify the bracket. Expanding A:

$$\begin{aligned} &\left(H_{12}^M + sH_{22}^M\right)^2 - \left(H_{11}^M + 2sH_{12}^M + s^2 H_{22}^M\right)H_{22}^M \\ &= \left(H_{12}^M\right)^2 + 2sH_{12}^M H_{22}^M + s^2\left(H_{22}^M\right)^2 - H_{11}^M H_{22}^M - 2sH_{12}^M H_{22}^M - s^2\left(H_{22}^M\right)^2 \\ &= \left(H_{12}^M\right)^2 - H_{11}^M H_{22}^M = -\det(\boldsymbol{H}_M). \end{aligned} \tag{73}$$

The s-linear and s-quadratic terms cancel pairwise, leaving just $-\det(\boldsymbol{H}_M)$, which is negative since $\boldsymbol{H}_M$ is positive definite.

Solve for d_0^2. Substituting the simplified bracket into Equation 72:

$$-d_0^2 \det(\boldsymbol{H}_M) = 2Af^* \quad \Rightarrow \quad d_0^2 = -2Af^* / \det(\boldsymbol{H}_M). \tag{74}$$

Both sides are positive: $f^* < 0$ at a real encounter minimum, $\det(\boldsymbol{H}_M) > 0$, and $A > 0$ (quadratic form of $\boldsymbol{H}_M$ evaluated along direction $(1, s)$).

Convert back to α. Using $d_0 = (2\pi/T_2)(\alpha - \alpha^*)$:

$$\Delta\alpha = (T_2/(2\pi))\sqrt{-2Af^*/\det(\boldsymbol{H}_M)}, \quad A = H_{11}^M + 2s\,H_{12}^M + s^2 H_{22}^M. \tag{75}$$

Interpretation. $\alpha = \alpha^*$: line through the minimum. $|\alpha - \alpha^*| = \Delta\alpha$: line tangent to the ellipse. $|\alpha - \alpha^*| > \Delta\alpha$: line misses it entirely, no SOI entry on this revolution pair. The admissibility band is shaded in Figure 3.

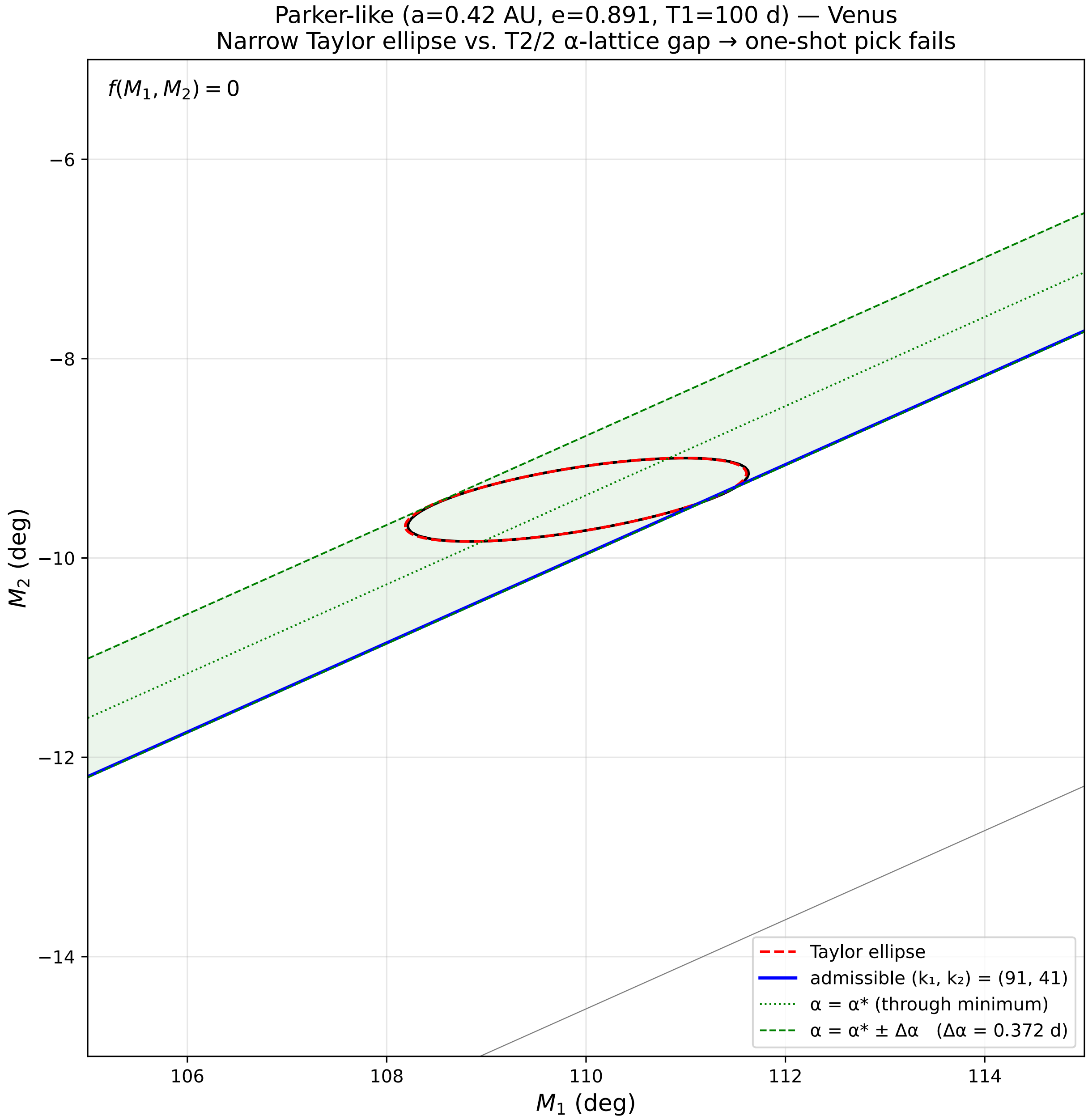

Figure 3: Parker-like spacecraft vs. Venus in the (M_1, M_2) mean-anomaly domain, cropped around the geometric minimum. The red dashed Taylor ellipse and the black $f = 0$ contour overlap to within line thickness. The green dotted line is $\alpha = \alpha^*$ (passing through the minimum); the dashed green lines at $\alpha = \alpha^* \pm \Delta\alpha$ are tangent to the Taylor ellipse and bound the admissibility band (shaded) derived in Section 7.2.1. Here $\Delta\alpha = 0.372$ days, very narrow relative to the gray-line spacing, so the admissible (k_1, k_2) is far from the nearest lattice line and is found by a continued-fraction search — the blue line is the first admissible pair, $(k_1, k_2) = (91, 41)$.

7.2.2. Continued-fraction search for admissible k_1

Given α^*, $\Delta\alpha$, and the mission-start floor $k_1^{\min}$ from Equation 67, the remaining task is: find the smallest integer $k_1 \geq k_1^{\min}$ such that Equation 66 produces a lattice point $\alpha := T_1 k_1 - T_2 k_2$ inside $[\alpha^* - \Delta\alpha, \alpha^* + \Delta\alpha]$. Define the residual

$$\rho(k_1) := T_1 k_1 - T_2 \cdot \mathrm{round}((T_1 k_1 - \alpha^*)/T_2) - \alpha^* = \alpha - \alpha^*. \tag{76}$$

By construction $\rho(k_1) \in (-T_2/2, T_2/2]$, and k_1 is admissible iff $|\rho(k_1)| \leq \Delta\alpha$. When $\Delta\alpha \geq T_2/2$ every k_1 is admissible and the answer is simply $k_1^{\min}$. The interesting case is $\Delta\alpha \ll T_2/2$, where admissible k_1 are sparse and must be located directly — without enumerating non-admissible values.

7.2.2.1. Key observation: CF convergents as calibrated jumps

As k_1 increases by 1, ρ advances by a single large step on the circle $\mathbb{R}/T_2\mathbb{Z}$. Stepping k_1 by the denominator q_n of the n -th CF convergent (p_n, q_n) of $\tau := T_1/T_2$ instead advances ρ by

$$\eta_n \cdot T_2 := q_n T_1 - p_n T_2, \tag{77}$$

which is **small** (alternating sign with n, shrinking roughly geometrically). Each convergent therefore gives a different calibrated step size in ρ-space.

Any non-negative integer Δk has a unique **Ostrowski representation**

$$\Delta k = \sum_{i \geq 0} c_i q_i, \quad 0 \leq c_{i+1} \leq a_{i+1}, \tag{78}$$

where the a_i are the partial quotients of $\tau = [a_0; a_1, a_2, ...]$. Stepping k_1 by Δk shifts ρ by $\sum_i c_i \cdot \eta_i T_2$. Choosing the c_i from large i (tiny shifts) down to small i (coarse shifts), we construct the smallest Δk that places ρ inside the admissibility band in $\mathcal{O}(\log(T_2/\Delta\alpha))$ steps.

7.2.2.2. Algorithm ($\mathcal{O}(\log)$)

Following Visser (2023) §§3.4 and 6.1, adapted to our notation:

1. Expand $\tau = T_1/T_2$ as a continued fraction,

$$\tau = [a_0; a_1, a_2, ...], \tag{79}$$

   and generate convergents (p_n, q_n) by the recursion

$$p_{-1} = 1,\ p_0 = a_0,\ p_{n+1} = a_{n+1} p_n + p_{n-1}, \quad q_{-1} = 0,\ q_0 = 1,\ q_{n+1} = a_{n+1} q_n + q_{n-1}. \tag{80}$$

   Continue until $|\eta_n| \cdot T_2 \leq \Delta\alpha$; call this depth L. Because q_n grows at least at the Fibonacci rate, $L = \mathcal{O}(\log(T_2/\Delta\alpha))$.

2. At each CF level $n = L, L-1, ..., 1$, test a **constant** number of integer candidates (typically 1–3) in the admissibility trapezoid at that level. The admissibility trapezoid is bounded by $((1-\delta)/q_n,\ (1+\delta)/q_n)$ along one basis direction, with the conjugate coordinate fixed by the ceiling rule of Equation 66. Each candidate is one integer multiplication plus a comparison.

3. Whenever a candidate at level n satisfies $|\rho| \leq \Delta\alpha$ **and** the corresponding $k_1 \geq k_1^{\min}$, translate it back to the standard basis,

$$k_1 = x \cdot k_{2n} - y \cdot k_{2n+1}, \tag{81}$$

   with (k_n) satisfying the same recursion as (q_n) but seeded by $(k_0, k_1) = (1, 0)$. Return this k_1.

4. If no candidate at any level satisfies both constraints, τ is rational (mean-motion resonance — the encounter does not recur at integer (k_1, k_2)) or the encounter is a geometric accident that cannot be converted into a real SOI crossing. Return failure.

Total cost: $\mathcal{O}(L) = \mathcal{O}(\log(T_2/\Delta\alpha))$ integer operations. The returned k_1 is the smallest admissible value.

Once k_1 is returned, the companion k_2 follows from Equation 66:

$$k_2 = \mathrm{round}((T_1 k_1 - \alpha^*)/T_2). \tag{82}$$

7.2.2.3. Worked example: Parker-like / Venus

With $\tau = T_1/T_2 = 0.447208$, the continued-fraction expansion is $\tau = [0; 2, 4, 4, 4, 12, 3, 1, ...]$. The first few convergents and their errors are:

n	p_n/q_n	$\lvert\eta_n\rvert \cdot T_2$ (s)	within $\Delta\alpha$?
1	$1/2$	2.05×10^6	no
2	$4/9$	4.83×10^5	no
3	$17/38$	1.18×10^5	no
4	$72/161$	9.64×10^3	**yes** — first time $\leq \Delta\alpha = 3.21 \times 10^4$ s

The Ostrowski construction terminates at depth $L = 4$, yielding $(k_1, k_2) = (91, 40)$ with $\rho(91) = -3.17 \times 10^4$ s $\in [-\Delta\alpha, \Delta\alpha]$. The corresponding time line (blue in Figure 3) crosses the Taylor ellipse, and the Newton refinement of Section 7.1 converges to the SOI entry point at $t_{\text{enc}} \approx 25.1$ years from epoch.

7.2.2.4. Worked example: Voyager 2 / Jupiter

With $\tau = 0.587689$, the CF expansion is $\tau = [0; 1, 1, 2, 2, 1, 5, ...]$. Here $\Delta\alpha = 59.06$ days is 2.7% of $T_2/2$ — wide enough that the admissibility condition $\lvert\rho(k_1^{\min})\rvert \leq \Delta\alpha$ is already satisfied at $k_1 = 0$. No Ostrowski expansion is needed: the algorithm returns $(k_1, k_2) = (0, 0)$ in one step, and the encounter occurs within the first revolution.

7.2.2.5. Refinement and final encounter time

After (k_1, k_2) is chosen, the specific time line

$$M_2 = (T_1/T_2)M_1 + c_{\text{line}}, \quad c_{\text{line}} = -(T_1/T_2)M_{1,0} + M_{2,0} + 2\pi\alpha/T_2 \tag{83}$$

is intersected with the Taylor ellipse (cf. Section 7.1) to obtain a seed E_1 at the SOI entry side of the ellipse, and a 1D Newton iteration along the line converges to the exact $f = 0$ crossing. The encounter time is then recovered via Kepler's equation applied to body 1:

$$t_{\text{enc}} = (T_1/(2\pi))\big(M_{1,\text{enc}} + 2\pi k_1 - M_{1,0}\big), \quad M_{1,\text{enc}} = E_1 - e_1 \sin E_1. \tag{84}$$

Equivalently via body 2 through the time-coupling constraint Equation 60; agreement to floating-point precision between the two is a useful invariant test.